



Stent Retriever Thrombectomy in Patients With Myocardial Infarction The Open-label, Multicenter NATURE Superiority trial

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Stent Retriever Thrombectomy in Patients With Myocardial Infarction The Open-label, Multicenter NATURE Superiority trial

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Running title: Stent retriever thrombectomy in myocardial infarction

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ABSTRACT

BACKGROUND Patients with myocardial infarction and large thrombus burden face greater infarct size and higher mortality, yet no effective therapy exists for this high-risk subgroup.

OBJECTIVES To investigate the effects of stent-retriever thrombectomy in addition to conventional percutaneous coronary intervention (PCI) versus conventional PCI alone on infarct size.

METHODS In this open-label, multicenter, superiority trial, 160 patients with ST-segment elevation myocardial infarction and large thrombus burden at angiography (TIMI thrombus grade ≥ 3 in infarct-related artery) were consented and 156 randomly assigned to undergo stent-retriever thrombectomy followed by conventional PCI or conventional PCI alone. Primary outcome was infarct size extension assessed by area under the curve (AUC) for creatine kinase (CK)-MB. Safety outcome was major adverse cardiovascular events (MACE; comprising cardiovascular death, nonfatal myocardial infarction, nonfatal stroke), at 3 and 30 days.

RESULTS Infarct size (CK-MB AUC) was lower in the stent-retriever group ($n=78$) versus the conventional group ($n=76$) (3965 IU/L·h; interquartile range [IQR] 2480–5092 versus 5250; IQR 3506–7449; difference -1359 , 95% CI -2231 to -522 ; $p=0.001$). At exploratory cardiac magnetic resonance findings, infarct size was lower in the stent-retriever group ($n=64$; left ventricle, 17%; IQR 11–28) versus the conventional PCI group ($n=59$; 28.0%; IQR 18–33; difference -7.0 , 95% CI -12 to -2.0), whereas left ventricular volumes and ejection fraction did not differ. There was no MACE at 3 days and 1 at 30 days in the stent-retriever group versus 2 and 3, respectively, in the conventional PCI group.

CONCLUSIONS Stent-retriever thrombectomy followed by conventional PCI in patients with large thrombus burden undergoing primary PCI ≤ 8 hours of symptom onset was associated with reduced infarct size versus conventional PCI alone.

This trial is registered with ClinicalTrials.gov (NCT04969471).

Condensed abstract

The NATURE trial investigated infarct size in 156 ST-segment elevation myocardial infarction patients with large thrombus burden who underwent percutaneous coronary intervention (PCI) ≤ 8 hours of symptom onset with/without stent-retriever-based thrombectomy by measuring AUC for creatine kinase (CK)-MB (primary endpoint). We investigated infarct size in 64 patients in the stent-retriever group and 59 patients in the conventional PCI group who underwent cardiac magnetic resonance imaging (MRI) at day 3. We showed that use of a stent retriever for performing thrombectomy before conventional PCI was associated with lower infarct size, based on both AUC of CK-MB and cardiac MRI.

KEY WORDS myocardial infarction – thrombectomy – percutaneous coronary intervention – stent retriever

Nonstandard Abbreviations and Acronyms

AUC	area under the curve
CK	creatine kinase
CK-MB	creatine kinase-MB isoform
MACE	major adverse cardiovascular events
MRI	magnetic resonance imaging
NATURE	eNvAst Thrombectomy device as Adjunctive Reperfusion Modality in the ST-segment Elevation Myocardial Infarction
PCI	percutaneous coronary intervention
STEMI	ST-segment elevation myocardial infarction
TIMI	Thrombolysis In Myocardial Infarction
hs-cTnT	High-Sensitivity Cardiac Troponin T

INTRODUCTION

In patients with myocardial infarction, the presence of a large coronary thrombus burden is associated with larger infarct size¹ and increased risk of major adverse cardiovascular events,^{2,3} including mortality,⁴ compared to patients without a large thrombus burden after primary percutaneous coronary intervention (PCI).

Manual aspiration catheters have been investigated as adjunctive treatments in patients with myocardial infarction regardless of thrombus burden.⁵⁻⁷ However, their use did not reduce infarct size⁸ or improve clinical outcomes⁹ compared with conventional PCI alone. In addition, concerns have been raised for a greater risk of stroke with manual aspiration, especially in patients with a large thrombus burden.⁹

The development of stent retriever technology constituted a paradigm shift in acute stroke intervention¹⁰ and led to a class I recommendation for endovascular thrombectomy in patients with acute large vessel occlusion.¹¹ The evidence for the use of stent retrievers in patients with myocardial infarction is encouraging but limited.^{12,13}

We conducted a multicenter randomized trial in patients with myocardial infarction and a large thrombus burden to compare stent-retriever thrombectomy in addition to conventional PCI to conventional PCI alone with respect to infarct size, left ventricular function, and clinical outcomes.

METHODS

STUDY DESIGN AND TRIAL OVERSIGHT. The NATURE (eNvAst Thrombectomy device as Adjunctive Reperfusion Modality in the ST-segment Elevation Myocardial Infarction) trial was a randomized, multicenter, open-label study designed to assess the superiority of the enVast stent retriever (Vesalio, Dallas, TX, USA) as an adjunctive reperfusion strategy in patients with a large thrombus burden undergoing primary PCI. The trial was designed by the first author, with

the support of the sponsor, an independent contract research organization and an independent statistician and was approved by the institutional review board at each centre.¹⁴

The first patient was randomized on December 10th, 2021. The last patient was included on January 29th, 2026. The database for the primary endpoint analysis was locked on June 1st 2026, while patient follow-ups remain ongoing.

PATIENTS. Patients with ST-segment elevation myocardial infarction (STEMI) within 8 hours from symptom onset were eligible if infarct-related artery showed a Thrombolysis In Myocardial Infarction (TIMI) thrombus grade ≥ 3 before PCI. In patients with a pre-PCI TIMI thrombus grade of 5, a TIMI thrombus grade ≥ 3 had to be confirmed by angiography after coronary wire insertion. Exclusion criteria included infarct-related artery diameter of less than 2.5 mm on visual assessment, heavily calcified plaque(s) or extreme vessel tortuosity proximal to or at the site of the culprit lesion(s), stent thrombosis as culprit lesion, and/or previous myocardial infarction in the same target vessel. A full list of inclusion and exclusion criteria is presented in the Supplementary Appendix. All patients gave written informed consent.

RANDOMIZATION. Patients were centrally randomized in a 1:1 ratio to receive open-label enVast-assisted mechanical thrombectomy followed by conventional PCI or conventional PCI alone. Randomization was concealed using a web-based system; randomization sequences were computer-generated, blocked, with randomly selected block sizes of 2 or 4, and stratified by site, time from symptom onset to intervention (≤ 3 versus > 3 hours), and intended use of parenteral antiplatelet drugs (i.e., glycoprotein IIb/IIIa inhibitors or cangrelor).¹⁴⁻¹⁶ Randomization was performed after angiography but prior to any intervention (except for lesion wiring if TIMI thrombus grade 5 was present).

PROCEDURES. Patients randomly allocated to the enVast-assisted mechanical thrombectomy group were treated with immediate enVast deployment in the occlusive/thrombotic coronary lesion followed by enVast retrieval under continuous aspiration achieved with three VacLok 60 ml vacuum pressure syringes connected at the side port of the guiding catheter hemostatic valve.

The enVast-thrombectomy is 6-french compatible device and was advanced at the site of occlusion or thrombus containing lesion with the use of a commercially available 0.21'' or 0.027'' microcatheter. The enVast-thrombectomy treatment sequence could be repeated at the discretion of the investigator, followed by conventional intervention, including balloon angioplasty or coronary stenting if deemed necessary by the treating interventional cardiologist. In the conventional arm, conventional PCI was to be pursued by the treating interventional cardiologist, including the use of balloons, stents, or a manual thrombus aspiration catheter. Detailed descriptions of the two treatment regimens have been published.¹⁴

OUTCOMES. The primary efficacy outcome was the infarct size extension assessed by the area under the curve (AUC) for the creatine kinase (CK)-MB isoform. Blood samples were obtained at admission, at 90 minutes, every 4 hours on day 1, and every 6 hours on days 2 and 3. CK-MB activity was performed using a standardized protocol and assay (Roche, COBAS® 6000 analyzer) either in a local laboratory or in a central core-lab (Theoreo, Italy), depending on assay availability at each recruiting site. Before enrollment, CK-MB AUC was prospectively designated as the primary endpoint in protocol version G, dated November 11, 2021, with the corresponding sample size change entered on ClinicalTrials.gov on November 17, 2021. The first 14 patients at the coordinating Swiss site were enrolled under the previously approved protocol version F, in which CK-MB AUC was a prespecified secondary endpoint, while subsequent enrollment occurred under locally approved version G or equivalent protocols. Secondary outcomes included AUC for total CK and high-sensitivity cardiac troponin T (hs-cTnT), cumulative ST-segment elevation resolution at standard 12-lead electrocardiogram, and cardiac magnetic resonance imaging (MRI) data at day 3 as previously described.¹⁴ Cardiac MRI data were quantified with dedicated software (QMass MR 7.5; Medis, Leiden, the Netherlands). Left ventricular volume, left ventricular mass, left ventricular ejection fraction, and the extent of oedema and necrosis were determined. Myocardial necrosis (defined as weight in grams of left ventricular tissue) was defined by the extent of abnormal delayed gadolinium enhancement,

whereas myocardium at risk (defined as weight in grams of left ventricular tissue) was defined by the extent of oedema (high-signal intensity on T2-weighted short T1 inversion-recovery images). Myocardial salvage was defined as the difference between myocardium at risk and myocardial necrosis normalized to myocardium at risk.

All MRI images and electrocardiograms were blinded-assessed by independent core laboratories (Supplementary Appendix). Safety endpoints included major adverse cardiovascular events (MACE) at day 3 and at day 30, defined as the composite of cardiovascular death, nonfatal myocardial infarction, or nonfatal stroke. Clinical outcomes were adjudicated by members who were unaware of trial-group assignment and included cardiovascular death, myocardial infarction, total stroke, bleeding, new-onset cardiogenic shock, rehospitalization for heart failure, stent thrombosis, and target vessel revascularization.

STATISTICAL ANALYSIS. The trial was designed to test whether stent-retriever thrombectomy followed by PCI, compared with conventional PCI alone, reduces infarct size, expressed as AUC for CK-MB. We calculated that 148 patients with the complete primary endpoint assessment were required to have a 90% chance of detecting, at a 5% significance level, a decrease in the primary outcome from 4000 IU/L·h in the control group, based on prior evidence,^{17,18} to 2800 in the experimental group.

Continuous variables are presented as median (interquartile range; IQR), and categorical variables as counts and percentages. The AUC for CK-MB, expressed in IU/L·h, was calculated for each subject using the trapezoidal method, and was compared using Wilcoxon–Mann–Whitney test. Between-group differences were estimated using the Hodges–Lehmann method. This estimate represents the median of all possible pairwise differences between participants in the two treatment groups and therefore may differ from the arithmetic difference between the two observed group medians. Adjustment for multiple comparisons was not prespecified and was not implemented for any secondary or exploratory endpoint. Secondary and exploratory endpoints are shown as point estimates of effects with 95% confidence intervals. The widths of

the intervals have not been adjusted for multiplicity, and the intervals could not be used in place of hypothesis testing.

ROLE OF THE FUNDING SOURCE. The trial was sponsored by Vesalio. The sponsor had no role in the statistical analysis, interpretation of the results, or manuscript preparation. The sponsor provided medical writing assistance for protocol development and exerted oversight of the CRO activities, as per legal European requirements, of the project management, clinical site monitoring, safety evaluation, and data management. Data collection and data monitoring was performed by a contract research organization (Supplementary Appendix). All authors vouch for the accuracy and completeness of the data and all analyses and for the fidelity of this report to the trial protocol.

RESULTS

PATIENT ENROLLMENT AND TREATMENT. From 10 December 2021 to 29 January 2026, 160 patients were consented across 11 sites in Switzerland, Italy, and the Netherlands, and 156 were randomized to stent-retriever thrombectomy with enVast (n=78) followed by conventional PCI or to conventional PCI alone (n=78) (Figure 1).

The baseline characteristics and angiographic data of the patients appeared to be well balanced between the treatment groups, except that the median systolic blood pressure was lower in patients in the stent-retriever group than in the conventional PCI group (Table 1). The median age was 62.5 years, 80.1% were men, 12.8% had diabetes, and 14.7% were in Killip class 2 or greater; 91.0% of patients received aspirin and 71.2% unfractionated heparin before intervention.

Almost all patients underwent radial access (Table 2). The infarct-related artery was the left anterior descending artery in 40.4%, 98.7% had TIMI thrombus grade 3 or more, 19.2% received cangrelor and 18.6% a glycoprotein IIb/IIIa inhibitor during intervention. After randomization, EnVast was deployed at least once in all patients in the stent-retriever group and in 1 patient in the conventional PCI group, whereas manual thrombectomy was used in 2 patients in the stent-

retriever group and in 31 patients in the conventional PCI group. Protocol deviations are shown in Supplemental Table 1. Central core-laboratory angiographic assessments are shown in Supplemental Table 2. There was no evidence of vessel dissection or perforation after use of the stent-retriever.

INFARCT SIZE. Infarct size estimation, based on CK-MB release in 78 patients in the stent-retriever group and 76 patients in the conventional PCI group, showed lower AUC with the stent retriever (3965.3 IU/L·h; interquartile range 2479.9 to 5092.4 vs 5250.3; interquartile range 3505.7–7448.5; difference –1358.9, 95% CI –2230.6 to –521.5; $p=0.001$) (Figure 2). The completeness of blood sampling at each time point is shown in Supplemental Table 3. Sensitivity analyses (Supplemental Table 4) and exploratory analyses according to prespecified subgroups (Supplemental Figure 1) showed consistent findings.

The median AUC for CK was 37,666.3 (22,949.9–68,276.1) IU/L·h in 78 stent-retriever patients and 45,371.3 (28,506.4–75,261.0) in 76 conventional PCI patients (difference –5699.4, 95% CI –16,584.0 to 4402.5). The median AUC for hs-cTnT was 138,704.8 ng/L·h (IQR 76,670–259,597; $n=78$) with the stent-retriever and 176,383.8 ng/L·h (IQR 99,193–256,123.1; $n=76$) with conventional PCI (difference –28,407.5, 95% CI –70,375 to 13,208.5). Median concentrations of CK-MB, CK, and hs-cTnT at each scheduled sampling time point are shown in Supplemental Table 5. AUCs of the biomarkers at 24 and 48 hours are shown in Supplemental Table 6.

A total of 64 patients in the stent-retriever group and 59 patients in the conventional PCI group underwent cardiac magnetic resonance imaging at day 3. Baseline characteristics of patients with or without MRI are reported in Supplemental Table 7. In exploratory analyses, infarct size was lower in the stent-retriever group (17%, interquartile range 11–28) compared with conventional PCI (28.0%; interquartile range 18–33; difference –7.0, 95% CI –12 to –2) (Figure 3 and Supplemental Table 8), whereas left ventricular volumes and ejection fraction were similar (Supplemental Table 8).

Exploratory matched 12-lead electrocardiogram analyses showed higher degree of complete and partial ST-segment elevation resolution in the stent-retriever group after PCI (Supplemental Table 9).

CLINICAL OUTCOMES. Clinical outcomes at 3 days and at 30 days did not differ in the two groups (Supplemental Tables 10 and 11). There was no MACE at 3 days and 1 at 30 days in the stent-retriever group versus 2 and 3, respectively, in the conventional PCI group.

DISCUSSION

In patients with myocardial infarction and a large thrombus burden at angiography, the use of a stent retriever for thrombectomy prior to PCI was associated with reduced infarct size, as reflected by lower CK-MB area-under-the-curve values (**Central Illustration**). Cardiac MRI suggested lower infarct size in the stent-retriever group and matched 12-lead electrocardiogram analyses were consistent with greater ST-segment elevation resolution.

We recruited patients with a large thrombus burden at angiography undergoing primary PCI for STEMI within 8 hours from symptom onset. Large thrombus burden is highly prevalent in STEMI undergoing primary PCI⁴ and patients with this feature have greater infarct size and higher mortality rates compared to patients without a large thrombus burden. We therefore hypothesized that early presenting patients with a large thrombus load would be most likely to benefit from thrombectomy.

No prior randomized study has assessed the benefit of thrombectomy exclusively in patients with myocardial infarction presenting relatively early and with a large thrombus burden. Current guidelines encompass the selective use of glycoprotein IIb/IIIa inhibitor treatment in patients with a large thrombus burden if bleeding risk is acceptable. However, no prior study has investigated the use of a glycoprotein IIb/IIIa inhibitor in this selected patient population.

A single-center study suggested improved indices of microvascular reperfusion (including angiographic blush grade and complete resolution of ST-segment elevation) and improved clinical outcomes (including mortality) with manual thrombectomy compared with conventional PCI.⁵ The study included all-comer patients with STEMI within 12 hours from symptom onset, irrespective of the presence of a large thrombus burden at angiography.

Two large-scale, randomized clinical trials evaluating routine manual aspiration versus conventional PCI alone in STEMI patients within 12 hours from symptom onset and irrespective of thrombus burden showed no benefit of this intervention on clinical outcomes.^{6,7} A safety concern emerged of an increased risk of stroke (particularly ischemic stroke) with manual aspiration, which was apparent within 48 hours of the index procedure.⁶ In a subsequent individual-patient data meta-analysis involving patients with a large thrombus burden, manual thrombus aspiration was potentially associated with lower risks of cardiovascular mortality but higher risk of stroke at 30 days.⁹

So far, no study has conclusively shown that manual aspiration reduces infarct size or effectively lowers thrombus burden compared with conventional PCI.¹⁴ Manual thrombectomy relies on contact aspiration, the effectiveness of which depends on lumen and clot sizes and clot composition: manual thrombectomy requires clot fragmentation and may expose to risk of distal or proximal embolization. A stent retriever is a self-expanding nitinol stent-like tubular structure that acts by entangling the thrombus to facilitate its removal. Although the use of stent retrievers is well established for the treatment of acute stroke,¹⁰ data for coronary intervention are limited. A prior single-center study, by Kotronias et al,¹³ assessed the use of the solitaire stent retriever in 27 patients versus manual thrombectomy or conventional PCI among individuals with a large thrombus burden presenting within 12 hours.

Unlike the study by Kotronias et al,¹³ we implemented a combined technique consisting of distal thrombus entrapment with the use of a stent retriever and proximal aspiration through the guiding catheter or the guiding catheter extension.^{12,14} This technique has been developed to

maximize safe and effective thrombus compartmentalization and removal, and mirrors the Solumbra and SAVE (Stent retriever Assisted Vacuum-locked Extraction) techniques, which are widely used for stroke intervention.¹⁹

Some limitations of our trial should be considered. Treatments were open label, reflecting the impossibility of masking interventions in a treatment-strategy trial. However, core-laboratory and outcome assessors were blinded to treatment assignment. This study is a proof-of-concept trial powered for infarct size based on CK-MB assessment. Infarct size with CK-MB assessment is highly correlated to infarct size at cardiac magnetic resonance imaging and mortality.²⁰ CK-MB assessment was associated with a ~10% increase in 90-day mortality per 1000-unit rise in the area under the time-concentration curve across various studies.²¹ Our exploratory secondary findings with cardiac magnetic resonance, although performed in a subset of patients (78.8%), supported the primary findings based on CK-MB analysis. Interventions affecting infarct size have a consequential effect on mortality.²² Our trial was too small to assess the role of lower infarct size on global systolic cardiac function, such as left ventricular ejection fraction or clinical outcomes. We followed a pragmatic approach to standardizing CK-MB activity assessment by centralizing to a core-laboratory only those sites where the assay of interest was not available locally. This study did not aim to include an all-comer population with STEMI: rather patients undergoing PCI within 8 hours from symptom onset with angiographic findings suggesting a large thrombus burden and suitable anatomy for effective and safe stent-retriever deployment (i.e., limited vessel calcification and tortuosity). Like almost all prior studies in STEMI, patients had to be able to provide consent (first oral followed by written consent after intervention) at the time of PCI which might have further impacted the external validity to the study to the broad MI population undergoing intervention. Our study randomized patients to the use of a stent-retriever as first intervention followed by conventional PCI versus conventional PCI alone. Conventional PCI in the two study arms was not protocol-mandated and allowed the use of manual aspiration and direct stenting at investigators' discretion, reflecting current

practice and guidelines recommendations. Manual aspiration was less frequent, and direct stenting was more frequent in the stent-retriever arm compared with conventional PCI alone. Therefore, the current study was not designed to assess the mechanistic role of stent retriever use in isolation versus conventional PCI.

In conclusion, in patients with a large thrombus burden undergoing primary PCI within 8 hours from symptom onset, stent-retriever thrombectomy, compared with conventional PCI, was associated with reduced infarct size.

Perspectives

Competency in Medical Knowledge:

In patients with myocardial infarction, the presence of a large coronary thrombus burden is associated with larger infarct size and increased risk of major adverse cardiovascular events, including mortality, after primary percutaneous coronary intervention (PCI). The use of manual aspiration catheters has not been associated with reduced infarct size or improved clinical outcomes compared with conventional PCI alone and has raised concerns for a greater risk of stroke.

Competency in Patient Care:

The use of a stent-retriever on infarct size in a total of 156 patients with ST-segment elevation myocardial infarction and a large thrombus burden who underwent PCI within 8 hours from symptom onset was associated with lower infarct size, as assessed based on both the AUC of CK-MB and Cardiac MRI.

Translational Outlook 1:

The use of a stent retriever should be investigated in future trials to assess whether thrombus burden mitigation before conventional PCI reduces morbidity and mortality in patients with STEMI and a large thrombus burden.

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Legends

Central Illustration. Study design, intervention, and results of the NATURE trial. *Abbreviations:* *AUC*, area under the curve; *CK*, creatine kinase; *hs-cTnT*, high-sensitivity cardiac troponin T; *IRA*, infarct-related artery; *MRI*, magnetic resonance imaging; *PCI*, percutaneous coronary intervention; *STEMI*, ST-segment elevation myocardial infarction; *TIMI*, Thrombolysis In Myocardial Infarction.

Figure 1. Consort diagram. *Abbreviations:* *AUC*, area under the curve; *CK*, creatine kinase; *MRI*, magnetic resonance imaging; *PCI*, percutaneous coronary intervention.

Figure 2. Assessment of infarct size by biomarker measurement. Concentration-time curves of biomarkers for infarct-size assessment during the first 72 hours after randomization are shown for the two treatment groups, with corresponding AUC values and between-group differences with 95% CIs. Error bars represent 95% confidence intervals at each time point. A table showing the number of available samples at each time point is displayed below each curve. Panel A shows CK-MB concentrations (IU/L) over time (primary endpoint). Panel B shows CK concentrations (IU/L) over time. Panel C shows hs-cTnT concentrations (ng/L) over time. *Abbreviations:* *AUC*, area under the curve; *CI*, confidence interval; *CK*, creatine kinase; *hs-cTnT*, high-sensitivity cardiac troponin T.

Figure 3. Cardiac magnetic resonance imaging at day 3. Infarct size assessment expressed as percentage of left ventricular mass (Panel A) and as grams of left ventricular tissue (Panel B) with late gadolinium enhancement. Panel A shows infarct size expressed as percentage of total left ventricle mass. Panel B shows infarct size expressed as grams of tissue. Horizontal lines indicate median values. *Abbreviations:* *LV*, left ventricular; *PCI*, percutaneous coronary intervention.

Table 1. Demographic and Clinical Characteristics of the Patients at Baseline

Characteristic	Stent-retriever (n=78)	Conventional PCI alone (n=78)
Age, years	62.5 (56.0, 71.0)	62.5 (55.0, 75.0)
Sex, male	63 (80.8%)	62 (79.5%)
Body-mass index (kg/m ²)	26.3 (23.5, 29.8)	27.2 (24.1, 30.8)
Diabetes mellitus	9 (11.5%)	11 (14.1%)
Arterial hypertension	39 (50.0%)	45 (57.7%)
Hyperlipidemia	40 (51.3%)	37 (47.4%)
Smoking status		
Current smoker	35 (44.9%)	35 (44.9%)
Previous smoker	11 (14.1%)	14 (17.9%)
Creatinine clearance <60 mL/min	9 (11.5%)	14 (17.9%)
Previous myocardial infarction		
None	76 (97.4%)	74 (94.9%)
One	1 (1.3%)	3 (3.8%)
Two or more	1 (1.3%)	1 (1.3%)
Previous PCI	3 (3.8%)	3 (3.8%)
Previous coronary artery bypass graft	1 (1.3%)	0 (0.0%)
Peripheral vascular disease	0 (0.0%)	4 (5.1%)
Previous cerebrovascular event	0 (0.0%)	3 (3.8%)
Chronic pulmonary disease	3 (3.8%)	2 (2.6%)
Congestive heart failure	1 (1.3%)	0 (0.0%)
Systolic blood pressure (mmHg)	124.0 (109.0, 136.0)	132.0 (115.0, 150.0)

Heart rate (beats per min)	75.0 (63.0, 87.0)	75.0 (65.0, 88.0)
Cardiac arrest at presentation	0 (0.0%)	1 (1.3%)
Killip class $\geq$ II	11 (14.1%)	12 (15.4%)
Symptom onset to randomization (min)	155.2 (109.0, 237.8)	170.2 (110.0, 263.4)
Symptom onset to randomization >3 h	30 (38.5%)	31 (39.7%)

Data are number (%) or median (Q1, Q3).

PCI, percutaneous coronary intervention.

Table 2. Investigator-reported Angiographic Characteristics and Medications

Characteristic	Stent-retriever (n=78)	Conventional PCI alone (n=78)	P value
Medications administered before the catheterization laboratory			
Aspirin	70 (89.7%)	72 (92.3%)	0.58
Clopidogrel	1 (1.3%)	2 (2.6%)	>0.999
Prasugrel	6 (7.7%)	4 (5.1%)	0.51
Ticagrelor	18 (23.1%)	12 (15.4%)	0.22
Unfractionated heparin	55 (70.5%)	56 (71.8%)	0.86
Unfractionated heparin dose (IU)	5000.0 (5000.0, 5000.0)	5000.0 (5000.0, 5000.0)	0.43
Medications administered in and after the catheterization laboratory			
Aspirin	7 (9.0%)	10 (12.8%)	0.44
Clopidogrel	1 (1.3%)	2 (2.6%)	>0.999
Prasugrel	6 (7.7%)	4 (5.1%)	0.51
Ticagrelor	40 (51.3%)	48 (61.5%)	0.20
Glycoprotein IIb/IIIa inhibitor	15 (19.2%)	14 (17.9%)	0.84
Cangrelor	16 (20.5%)	14 (17.9%)	0.69
Unfractionated heparin	77 (98.7%)	75 (96.2%)	0.62
Unfractionated heparin dose	8000.0 (7000.0, 10,000.0)	7500.0 (6000.0, 8,500.0)	0.10
Arterial access site			
Radial access	78 (100.0%)	73 (93.6%)	0.06

Vessel disease			
Single vessel	37 (47.4%)	41 (52.6%)	0.52
Two vessels	23 (29.5%)	22 (28.2%)	0.86
Three vessels	18 (23.1%)	12 (15.4%)	0.22
Infarct-related artery			
Left main coronary	0 (0.0%)	1 (1.3%)	>0.999
Left anterior descending	31 (39.7%)	32 (41.0%)	0.87
Left circumflex	13 (16.7%)	12 (15.4%)	0.83
Right coronary artery	34 (43.6%)	35 (44.9%)	0.87
Bypass graft	0 (0.0%)	0 (0.0%)	
TIMI flow grade in the infarct-related artery			0.22
0	58 (74.4%)	55 (70.5%)	
1	4 (5.1%)	6 (7.7%)	
2	12 (15.4%)	7 (9.0%)	
3	4 (5.1%)	10 (12.8%)	
Initial TIMI thrombus grade			>0.999
1	1 (1.3%)	1 (1.3%)	
3	9 (11.5%)	9 (11.5%)	
4	13 (16.7%)	13 (16.7%)	
5	55 (70.5%)	55 (70.5%)	
TIMI thrombus grade after wire insertion			0.33
0	1/55 (1.8%)	3/56 (5.4%)	
1	0/55 (0.0%)	3/56 (5.4%)	

2	1/55 (1.8%)	2/56 (3.6%)	
3	13/55 (23.6%)	14/56 (25.0%)	
4	24/55 (43.6%)	16/56 (28.6%)	
5	16/55 (29.1%)	18/56 (32.1%)	
EnVast deployed at least once	78 (100.0%)	1 (1.3%)	<0.001
EnVast deployed twice or more	18 (23.1%)	0 (0.0%)	<0.001
Intermediate catheter used with stent-retriever	35/75 (46.7%)	0/1 (0.0%)	<0.001
Concomitant manual aspiration during stent-retriever use	60/75 (80.0%)	0/1 (0.0%)	<0.001
Device delivery rate	78 (100.0%)	1 (100.0%)	>0.999
Manual thrombectomy	2 (2.6%)	31 (39.7%)	<0.001
At least one stent implanted	75 (96.2%)	77 (98.7%)	0.62
Number of stents	1.0 (1.0, 2.0)	1.0 (1.0, 2.0)	0.34
Total stent length per patient (mm)	36.0 (24.0, 51.5)	36.0 (23.0, 48.0)	0.66
Direct stenting (no balloon predilatation)*	51/75 (68.0%)	24/77 (31.2%)	<0.001
Post-dilatation	39/75 (52.0%)	51/77 (66.2%)	0.07
At least two vessels treated	5/76 (6.6%)	3 (3.8%)	0.49
One lesion	52/76 (68.4%)	56 (71.8%)	0.65
Two lesions	18/76 (23.7%)	15 (19.2%)	0.50
Three or more lesions	6/76 (7.9%)	7 (9.0%)	0.81
Number of lesions with PCI	1.0 (1.0, 1.0)	1.0 (1.0, 1.0)	0.14
At least one lesion stented	75 (96.2%)	77 (98.7%)	0.62
At least one lesion not stented	7 (9.0%)	2 (2.6%)	0.17
Total stent length per lesion (mm)	30.0 (21.0, 48.0)	34.0 (23.0, 48.0)	0.61
Average stent diameter per lesion (mm)	3.5 (3.0, 3.9)	3.5 (3.0, 3.8)	0.20

Intra-aortic balloon pump	0 (0.0%)	0 (0.0%)	
Microaxial blood pump (Impella)	0 (0.0%)	1 (1.3%)	>0.999
Extracorporeal membrane oxygenation	0 (0.0%)	0 (0.0%)	
Slow or no flow treatment	16 (20.5%)	15 (19.2%)	0.84
Adenosine	7 (9.0%)	7 (9.0%)	>0.999
Nitroprusside	8 (10.3%)	5 (6.4%)	0.385
Inotropes	4 (5.1%)	6 (7.7%)	0.513
Total amount of contrast (ml)	150.0 (100.0, 200.0)	140.0 (100.0, 180.0)	0.37
Fluoroscopy time (sec)	698.0 (488.0, 989.0)	583.0 (426.0, 860.0)	0.11

Data are number (%) or median (Q1, Q3). The denominator is 78 for each study group

percentage unless otherwise reported.

* Among patients undergoing manual thrombectomy, direct stenting was performed in 2/2 (100.0%) in the Stent-retriever group and 9/31 (29.0%) in the conventional PCI group.

PCI, percutaneous coronary intervention; TIMI, Thrombolysis in Myocardial Infarction.

FIGURES

Central Illustration. Study design, intervention, and results of the NATURE trial.

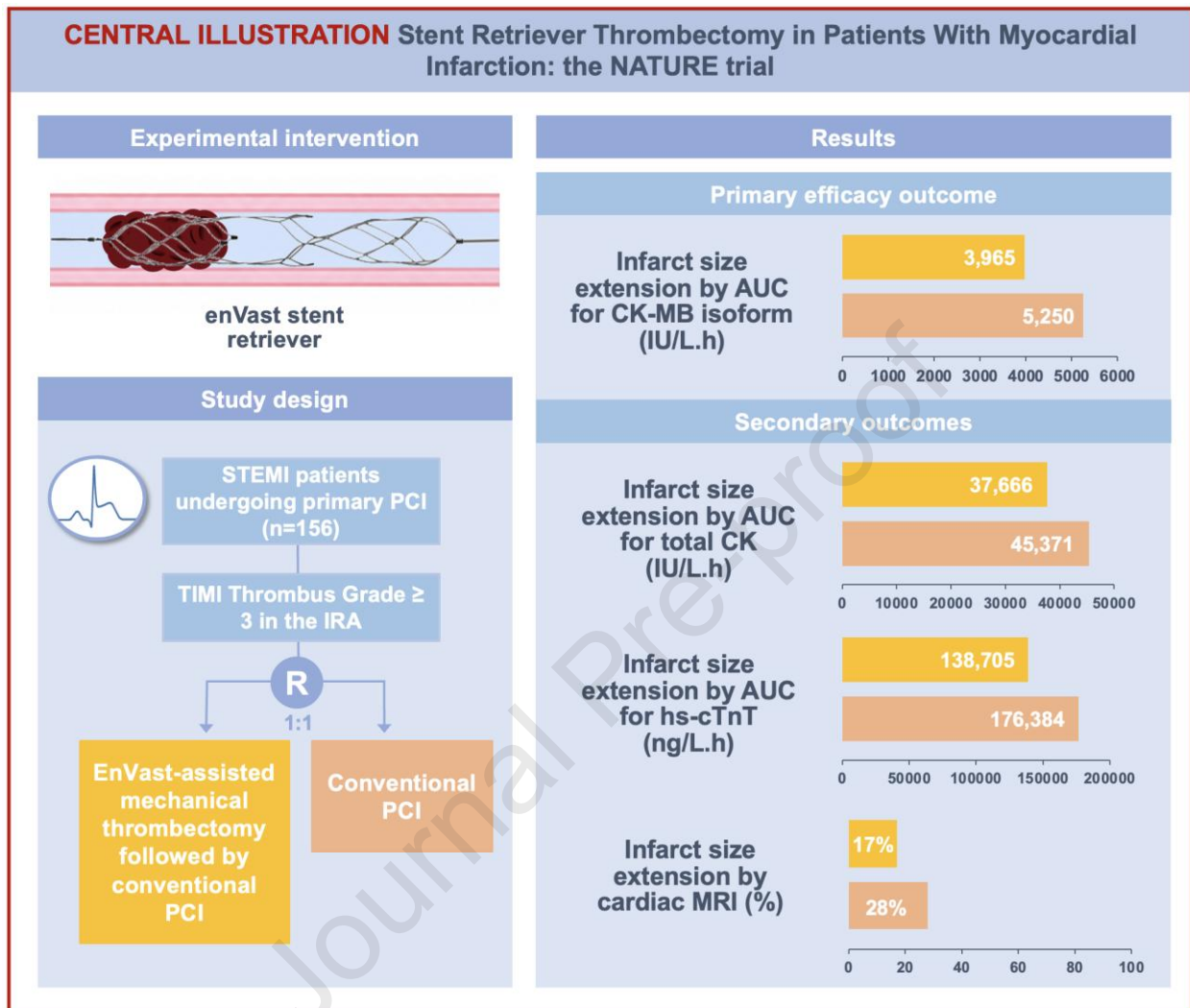


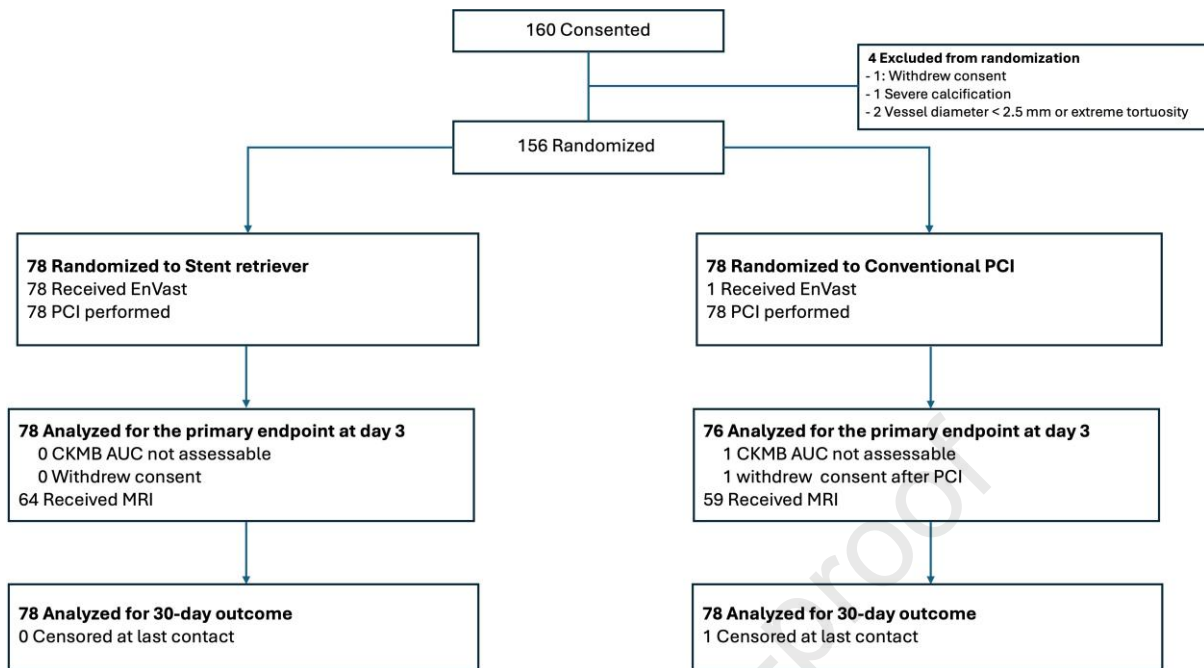
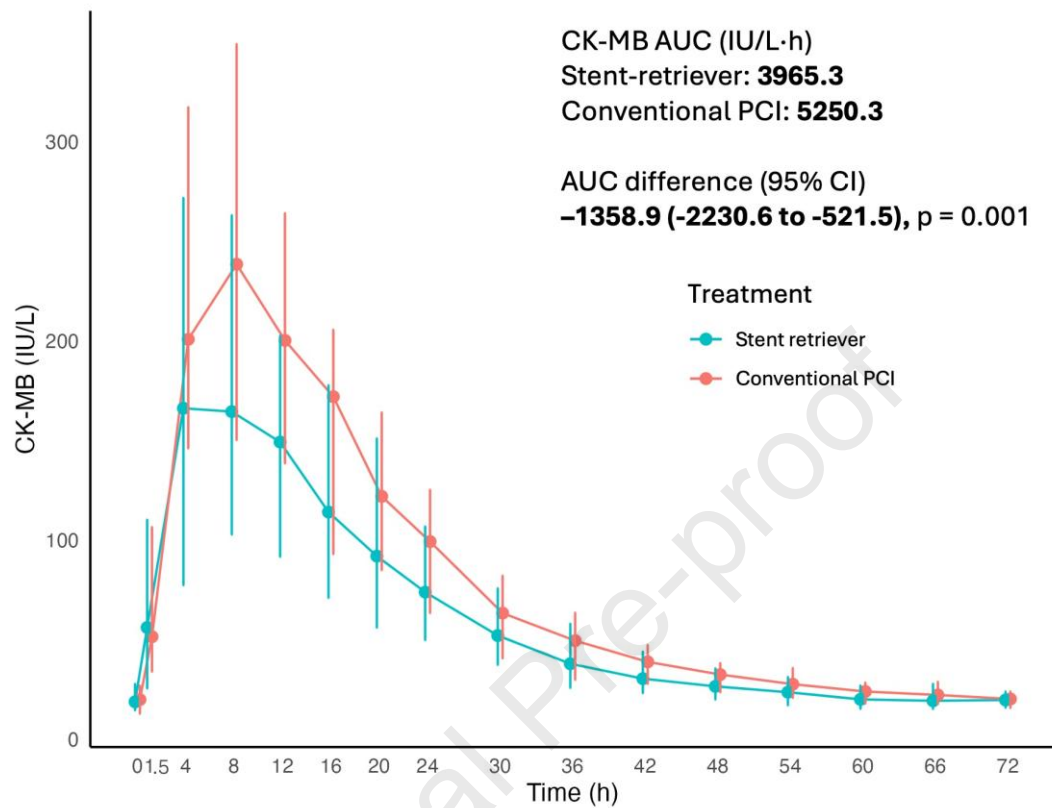
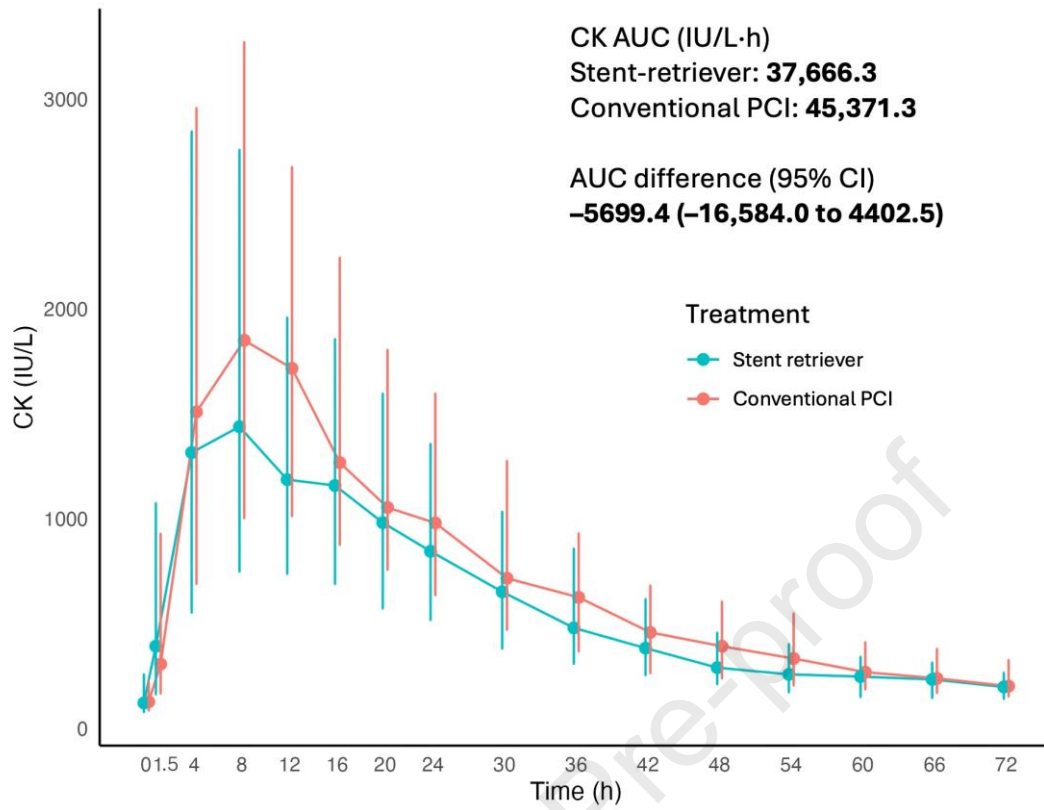
Figure 1. Consort diagram

Figure 2. Assessment of infarct size by biomarker measurement**Figure 2A**

	0 h	1.5 h	4 h	8 h	12 h	16 h	20 h	24 h	30 h	36 h	42 h	48 h	54 h	60 h	66 h	72 h
Stent-retriever (n=78)	69	66	66	74	72	66	73	67	66	69	66	68	62	60	54	53
Conventional PCI (n=78)	65	71	73	74	70	70	69	66	63	67	64	60	56	58	58	57

Figure 2B

	0 h	1.5 h	4 h	8 h	12 h	16 h	20 h	24 h	30 h	36 h	42 h	48 h	54 h	60 h	66 h	72 h
Stent-retriever (n=78)	70	66	67	73	70	65	69	66	66	68	65	64	58	58	55	52
Conventional PCI (n=78)	66	69	72	71	70	68	69	64	61	64	62	59	56	58	57	57

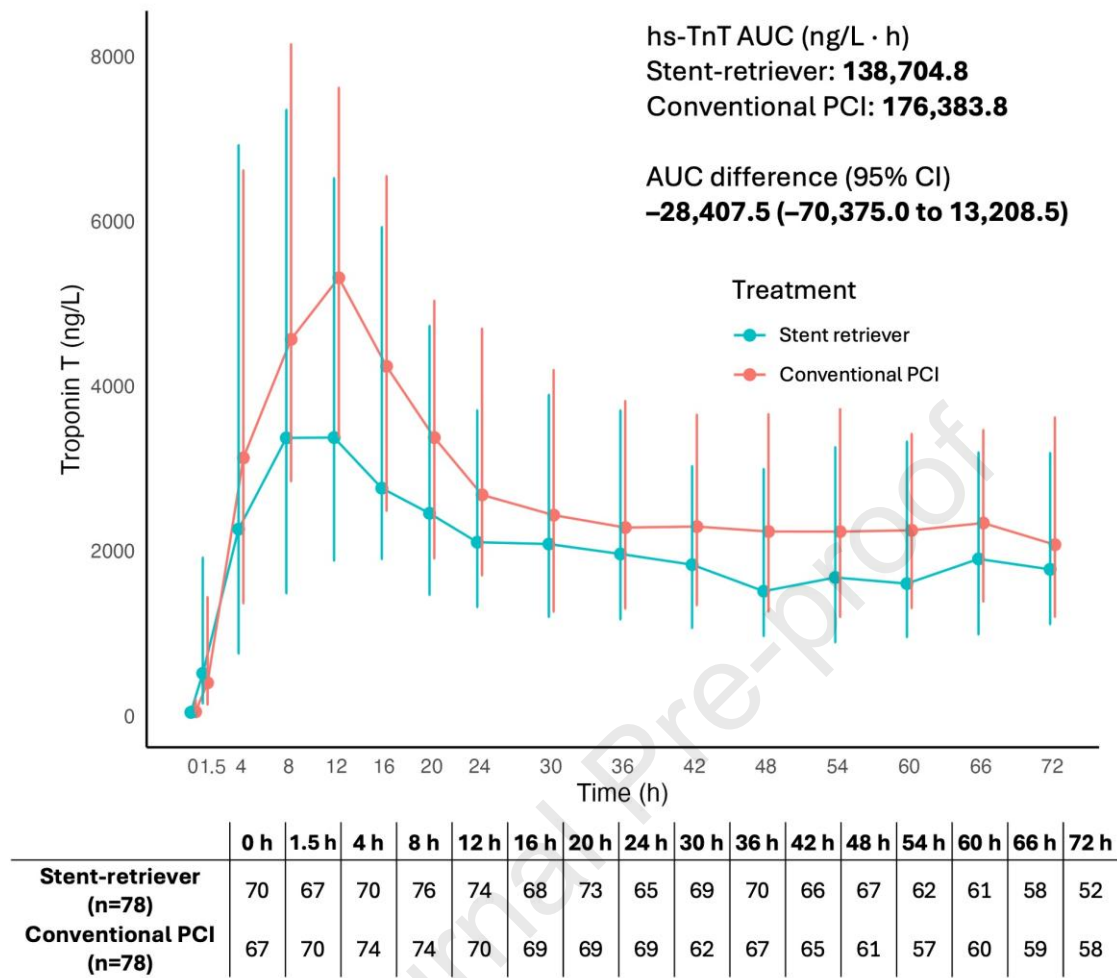
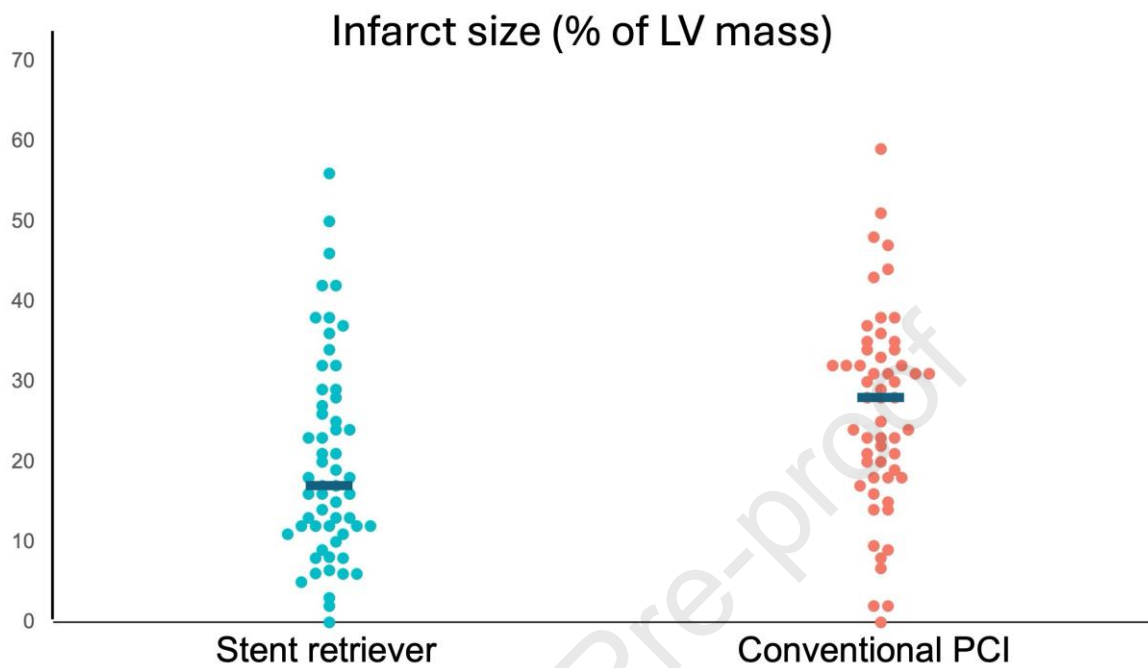
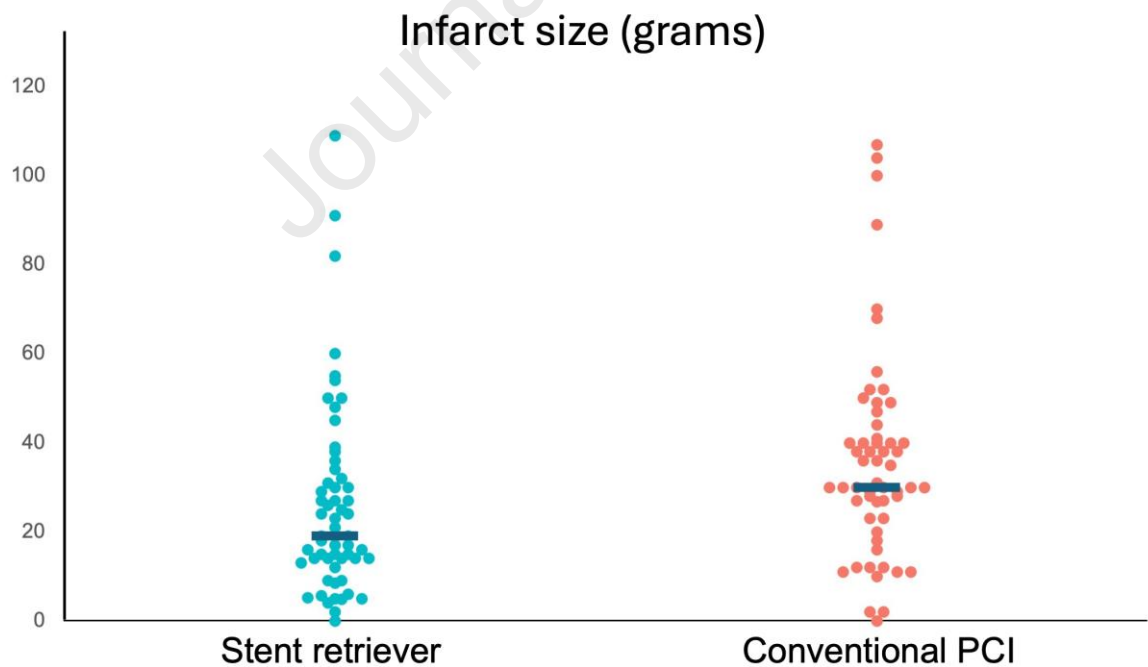
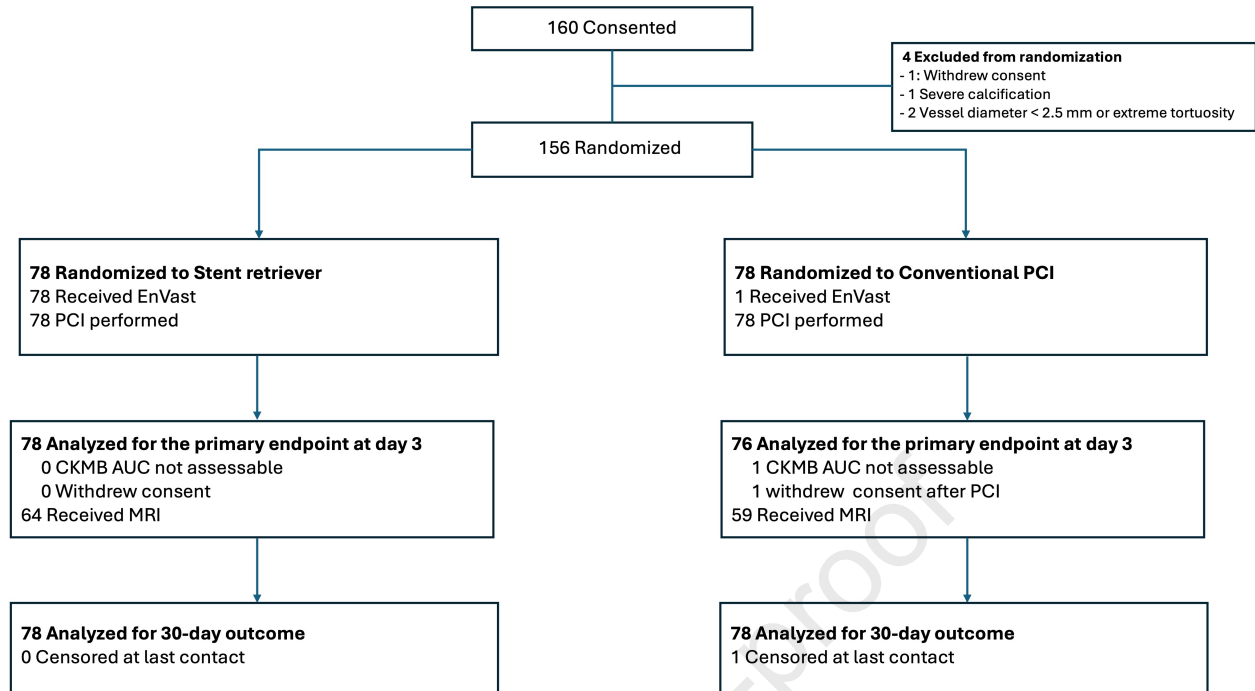
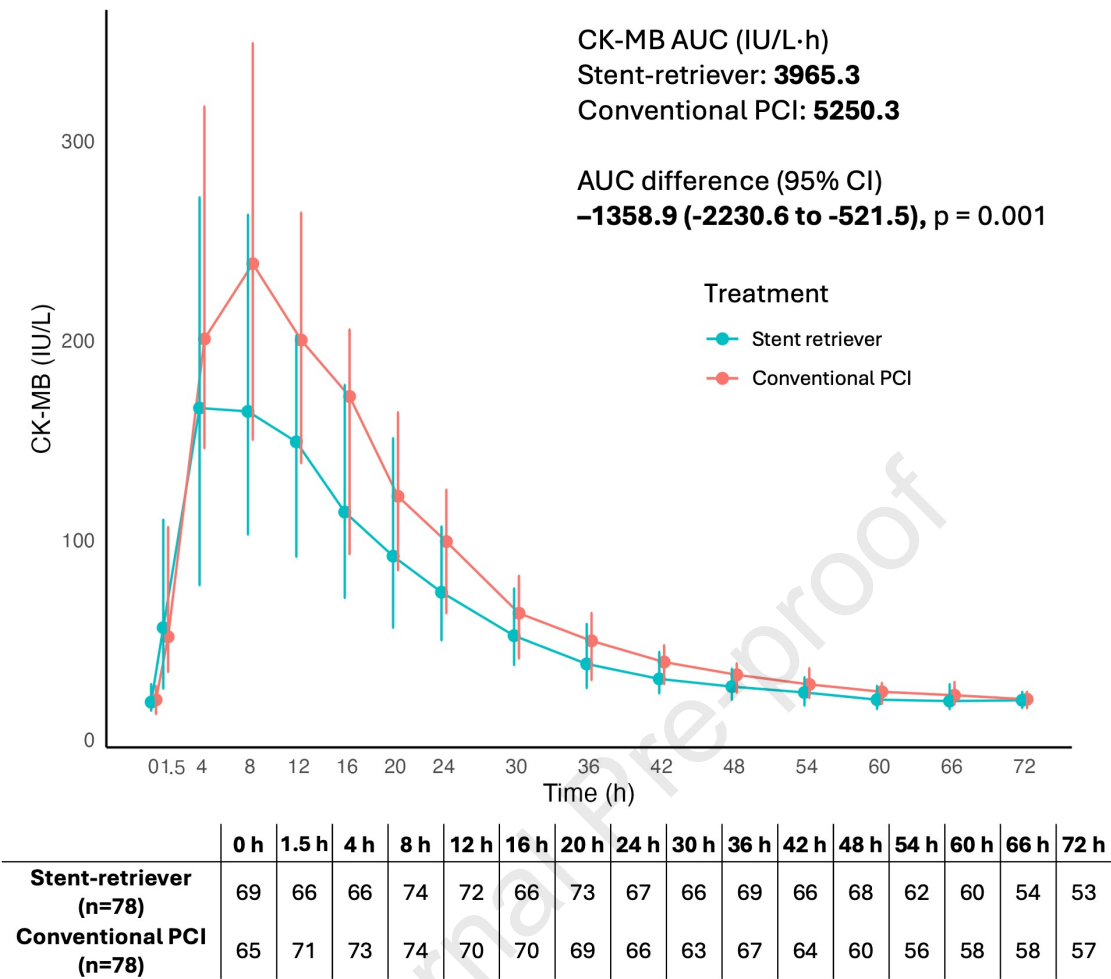
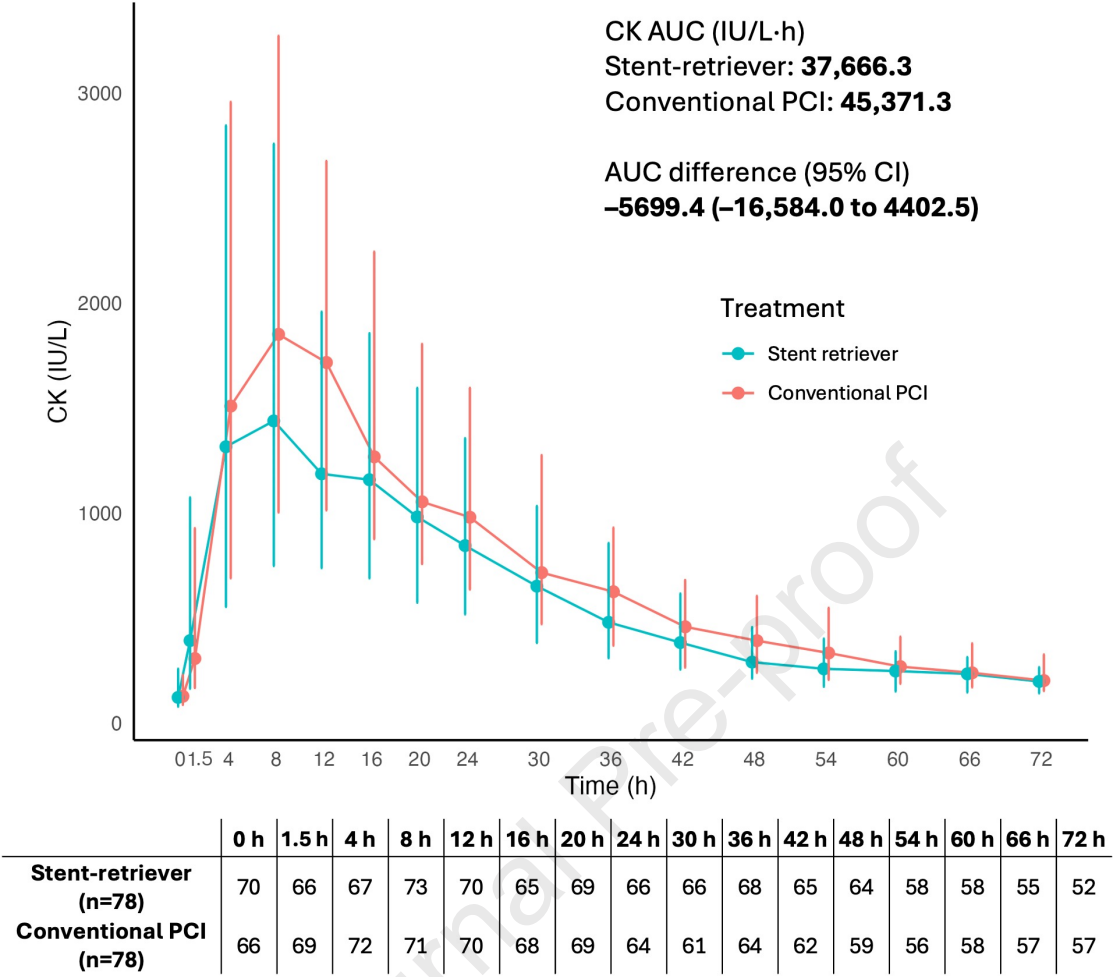
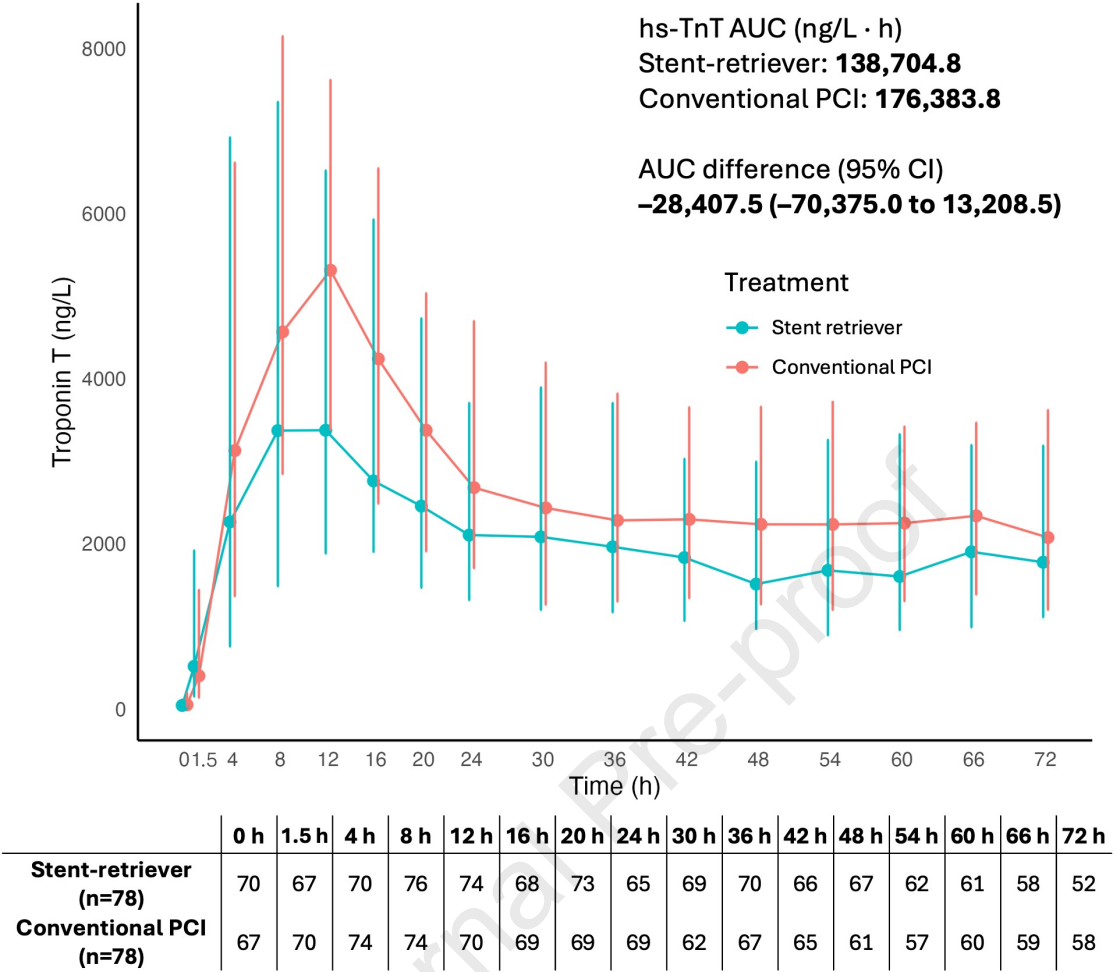
Figure 2C

Figure 3. Cardiac magnetic resonance imaging at day 3**Panel A****Panel B**

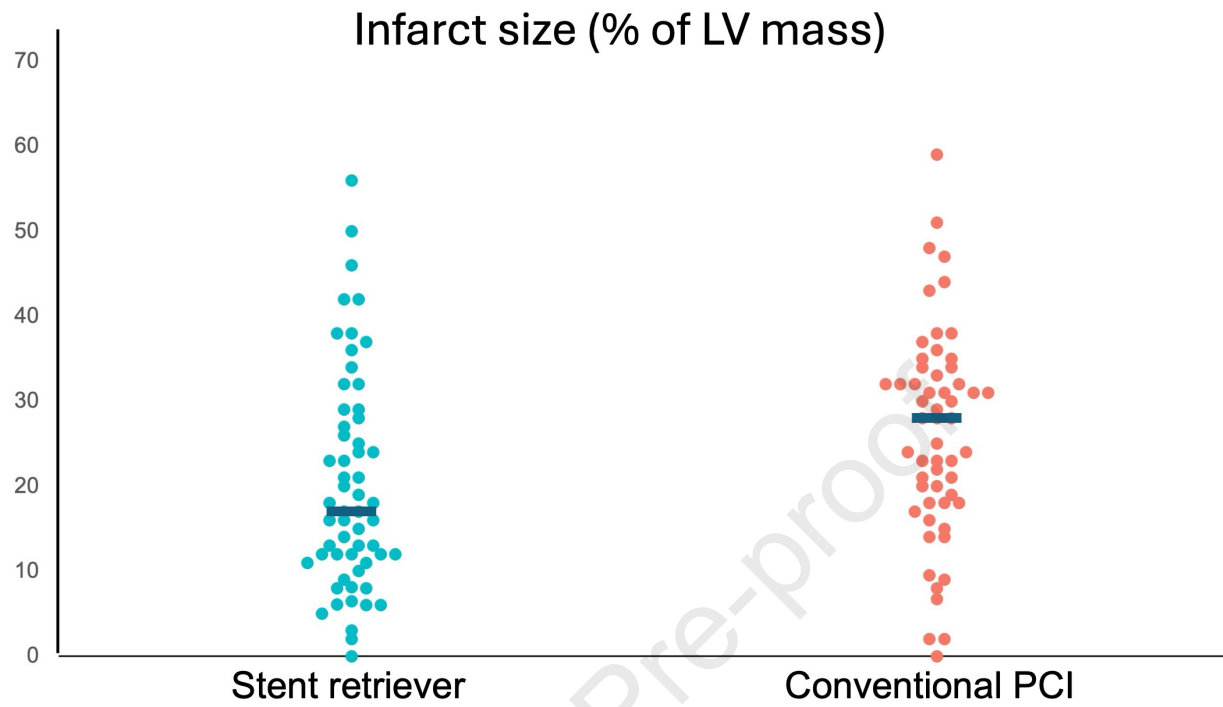




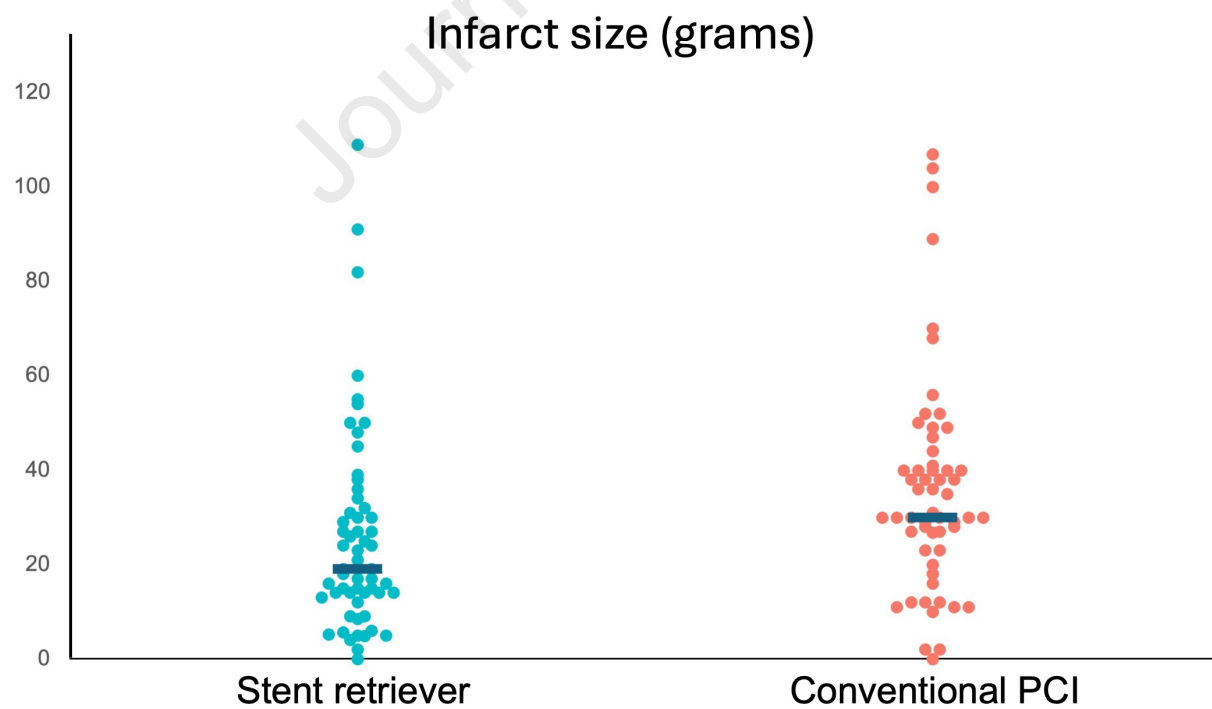




Panel A



Panel B



CENTRAL ILLUSTRATION Stent Retriever Thrombectomy in Patients With Myocardial Infarction: the NATURE trial